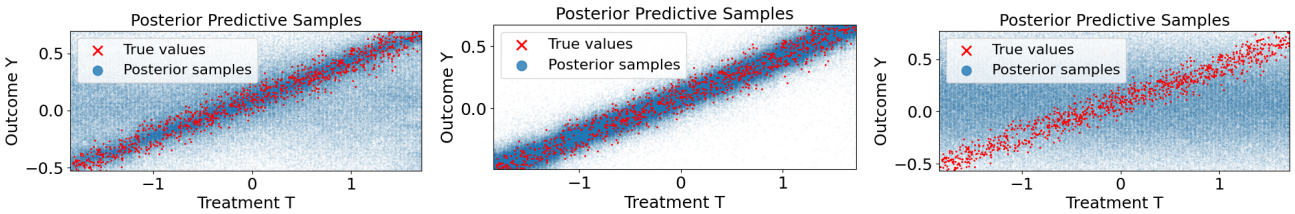

Figure 1. Here, we qualitatively demonstrate what happens when providing a Causal Foundation Model (CFM) with incorrect graph information in the unidentifiable bivariate case of a linear mechanism and Gaussian noise. As expected, since the data does not provide any evidence regarding the causal direction, the model, while being correct under the wrong assumption, outputs the "wrong" posterior: More specifically, this plot shows posterior predictive samples in the two-node case where T causes Y ($T \rightarrow Y$): i.e., $Y = 0.25 \cdot T + \epsilon$, for $\epsilon \sim \mathcal{N}(0, 0.1)$. Here, the causal direction is not identifiable from observational data. **Left**: Without telling the model Q_θ trained on arbitrary mechanisms (as, e.g., in DoPFN) the correct causal direction, it outputs a mixture distribution $Q_\theta(Y|\text{do}(T), T \overset{?}{\longleftrightarrow} Y) = 0.5 \cdot P(Y|\text{do}(T), Y \rightarrow T) + 0.5 \cdot P(Y|\text{do}(T), T \rightarrow Y)$. **Middle**: When conditioned on the right causal graph, the output aligns with the correct causal effect: $Q_\theta(Y|\text{do}(T), T \rightarrow Y) = P(Y|\text{do}(T), T \rightarrow Y)$. **Right:** When provided with the wrong causal information $T \leftarrow Y$, the model makes the prediction $Q_\theta(Y|\text{do}(T), T \leftarrow Y) = P(Y|\text{do}(T), T \leftarrow Y)$, which is the correct thing to do under the wrong assumption $T \leftarrow Y$.